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2.1 Simple execution, without data forwarding techniques

e)

Clock cycles	308
Instructions	170
Average CPI	1,81

Stalls:	- Data	104
	- Structural	0
Flushes:	- Branch Taken	15

f) Ao analisar a execução do programa, verificamos que, em todas as iterações do ciclo, são iniciadas pipelines para as instruções addi e ecall no final do código (ou seja, é feito o fetch de ambas). No entanto, na maioria das vezes essas instruções acabam por ser descartadas devido a um flush da pipeline. Desta forma, concluímos que o simulador adota a política de branch not taken, assumindo, em cada iteração, que o salto não ocorre, ou seja, que a iteração atual será a última do ciclo.

2.2 Application of data forwarding techniques

c)

Clock cycles	222
Instructions	170
Average CPI	1,31

Stalls:	- Data	18
	- Structural	0
Flushes:	- Branch Taken	15

d)

$$Speedup = \frac{CC_{2.1}}{CC_{2.2}} = \frac{308}{222} = 1,387$$

2.3 Source code optimization: minimization of data and structural hazards

a) Attach a copy of the new assembly program.

c)

Clock cycles	206
Instructions	170
Average CPI	1,21

Stalls:	- Data	2
	- Structural	0
Flushes:	- Branch Taken	15

d) Segundo a lógica anterior:

$$Speedup = \frac{CC_{2.1}}{CC_{2.3}} = \frac{308}{206} = 1,495$$

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2.4 Source code optimization: loop unrolling

a) Attach a copy of the new assembly program.

c)

Clock cycles	122
Instructions	110
Average CPI	1,11

Stalls:	- Data	2
	- Structural	0
Flushes:	- Branch Taken	3

d)

Segundo a lógica anterior:

$$\text{Speedup} = \frac{CC_{2.1}}{CC_{2.4}} = \frac{308}{122} = 2{,}525$$

Código desenvolvido:

2.3.

Source code

```
.data
A:  .word 1,2,3,4,5,6,7,8,9
    .word 10,11,12,13,14,15,16
B:  .word 11,22,33,44,55,66,77
    .word 88,99,111,122,133
    .word 144,155,166
C:  .word 0,0,0,0,0,0,0,0,0
    .word 0,0,0,0,0,0,0

    .text
main:
    addi s1, zero, 0   # i = 0
    addi s2, zero, 16  # value of N
    la s3, A           # base of A
    la s4, B           # base of B
    la s5, C           # base of C

loop:
    lw t1, 0(s3)
    lw t2, 0(s4)
    addi s1, s1, 1
    mul t3, t2, t1
    addi s3, s3, 4
    add t3, t3, t1
    addi s4, s4, 4
    sw t3, 0(s5)
    addi s5, s5, 4
    bne s1, s2, loop

end:
    addi a7, zero, 10
    ecall             # Exit (syscall)
```

2.4.

Source code

```
.data
A:  .word 1,2,3,4,5,6,7,8,9
    .word 10,11,12,13,14,15,16
B:  .word 11,22,33,44,55,66,77
    .word 88,99,111,122,133
    .word 144,155,166
C:  .word 0,0,0,0,0,0,0,0,0
    .word 0,0,0,0,0,0,0

    .text
main:
    addi s1, zero, 0   # i = 0
    addi s2, zero, 16  # value of N
    la s3, A           # base of A
    la s4, B           # base of B
    la s5, C           # base of C

loop:
    #1
    lw t1, 0(s3)
    lw t2, 0(s4)
    addi s1, s1, 4
    mul t3, t2, t1
    add t3, t3, t1

    #2
    lw t4, 4(s3)
    lw t5, 4(s4)
    sw t3, 0(s5)   # do loop anterior
    mul t6, t5, t4
    add t6, t6, t4

    #3
    lw t0, 8(s3)
    lw a1, 8(s4)
    sw t6, 4(s5)    # do loop anterior
    mul a2, a1, t0
    add a2, a2, t0

    #4
    lw a3, 12(s3)
    lw a4, 12(s4)
    sw a2, 8(s5)    # do loop anterior
    mul a5, a4, a3
    add a5, a5, a3
    sw a5, 12(s5)


    addi s3, s3, 16
    addi s4, s4, 16
    addi s5, s5, 16
    bne s1, s2, loop

end:
    addi a7, zero, 10
    ecall             # Exit (syscall)
```


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Table 1: Pipeline time diagram, without data forwarding techniques.

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	X	M	W														F																					
2	lw x7 0 x20		F	D	X	M	W																																		
3	mul x28 x7 x6			F	D	–	–	X	M	W																															
4	add x28 x28 x6				F	–	–	D	–	–	X	M	W																												
5	sw x28 0 x21							F	–	–	D	–	–	X	M	W																									
6	addi x9 x9 1										F	–	–	D	X	M	W																								
7	addi x19 x19 4													F	D	X	M	W																							
8	addi x20 x20 4														F	D	X	M	W																						
9	addi x21 x21 4															F	D	X	M	W																					
10	bne x9 x18 -36 <loop>																F	D	X	M	W																				
11	addi x17 x0 10																	F	D																						
12	ecall																		F																						
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Table 2: Pipeline time diagram, with data forwarding techniques.

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	X	M	W									F																										
2	lw x7 0 x20		F	D	X	M	W																																		
3	mul x28 x7 x6			F	D	–	X	M	W																																
4	add x28 x28 x6				F	–	D	X	M	W																															
5	sw x28 0 x21						F	D	X	M	W																														
6	addi x9 x9 1							F	D	X	M	W																													
7	addi x19 x19 4								F	D	X	M	W																												
8	addi x20 x20 4									F	D	X	M	W																											
9	addi x21 x21 4										F	D	X	M	W																										
10	bne x9 x18 -36 <loop>											F	D	X	M	W																									
11	addi x17 x0 10												F	D																											
12	ecall													F																											
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Table 3: Pipeline time diagram, with minimization techniques to reduce the data and structural hazards.

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	X	M	W								F																											
2	lw x7 0 x20		F	D	X	M	W																																		
3	addi x9 x9 1			F	D	X	M	W																																	
4	mul x28 x7 x6				F	D	X	M	W																																
5	addi x19 x19 4					F	D	X	M	W																															
6	add x28 x28 x6						F	D	X	M	W																														
7	addi x20 x20 4							F	D	X	M	W																													
8	sw x28 0 x21								F	D	X	M	W																												
9	addi x21 x21 4									F	D	X	M	W																											
10	bne x9 x18 -36 <loop>										F	D	X	M	W																										
11	addi x17 x0 10											F	D																												
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Table 4: Pipeline time diagram: usage of loop unrolling minimization techniques to reduce the control hazards.

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	X	M	W																							F												
2	lw x7 0 x20		F	D	X	M	W																																		
3	addi x9 x9 4			F	D	X	M	W																																	
4	mul x28 x7 x6				F	D	X	M	W																																
5	add x28 x28 x6					F	D	X	M	W																															
6	lw x29 4 x19						F	D	X	M	W																														
7	lw x30 4 x20							F	D	X	M	W																													
8	sw x28 0 x21								F	D	X	M	W																												
9	mul x31 x30 x29									F	D	X	M	W																											
10	add x31 x31 x29										F	D	X	M	W																										
11	lw x5 8 x19											F	D	X	M	W																									
12	lw x11 8 x20												F	D	X	M	W																								
13	sw x31 4 x21													F	D	X	M	W																							
14	mul x12 x11 x5														F	D	X	M	W																						
15	add x12 x12 x5															F	D	X	M	W																					
16	lw x13 12 x19																F	D	X	M	W																				
17	lw x14 12 x20																	F	D	X	M	W																			
18	sw x12 8 x21																		F	D	X	M	W																		
19	mul x15 x14 x13																			F	D	X	M	W																	
20	add x15 x15 x13																				F	D	X	M	W																
21	sw x15 12 x21																					F	D	X	M	W															
22	addi x19 x19 16																						F	D	X	M	W														
23	addi x20 x20 16																							F	D	X	M	W													
24	addi x21 x21 16																								F	D	X	M	W												
25	bne x9 x18 -96 <loop>																									F	D	X	M	W											
26	addi x17 x0 10																										F	D													
27	ecall																											F													
28																																									
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